

MeasureSure

SIH26036 | Online Verification System for Weighing and Measuring Instruments

Project brief and implementation concept | 29 September 2026

MeasureSure is a secure, role-based digital portal that helps citizens, businesses, and Legal Metrology inspectors register, verify, monitor, and renew weighing and measuring instruments. The product is designed around a simple public promise: every measurement should be traceable to a current, trusted verification record.

1. Problem statement

Verification records are often spread across paper certificates, local registers, and disconnected spreadsheets. This makes it difficult to confirm whether an instrument is valid at the point of use, creates avoidable work for inspectors, and increases the risk of expired or tampered certificates being accepted.

2. Proposed solution

MeasureSure creates one digital record per instrument. Each record carries a unique certificate ID, owner and location details, instrument type, verification history, renewal date, and a digitally signed status. The public search experience is intentionally lightweight: a user can search by certificate number, serial number, or owner name and immediately see whether the instrument is valid, due soon, or pending review.

3. Primary users and journeys

User	Primary journey	Outcome
Citizen / buyer	Search certificate before purchase or transaction	Quick confidence in the reading
Business owner	Register instrument and track renewal due dates	Fewer missed renewals
Inspector	Review queue, verify on site, issue certificate	Faster, auditable field work
Department admin	Monitor district activity and exceptions	Clear operational visibility

4. Product scope

- Public verification search with certificate, serial, and owner lookup.
- Instrument registration with owner, location, type, and supporting evidence.
- Inspector workspace with inspection queue, verification actions, and certificate issuance.
- Certificate repository with status, audit trail, renewal dates, and downloadable records.
- Dashboard metrics for registered, verified, due-soon, and pending instruments.
- Tamper-evident digital signatures and role-based access control.

5. Prototype interface

The delivered website prototype uses a calm government-service visual language: a clear left navigation, a public verification panel, a compact operational dashboard, and a recent-records table. The main interaction is live in the

browser: type a certificate or owner query to filter the record set, use the navigation to change workspace context, or trigger registration and record actions.

6. Suggested technical architecture

Layer	Recommended implementation
Frontend	Next.js App Router, responsive React components, accessible form controls
API	REST or typed RPC endpoints for registration, lookup, verification, and renewal
Data	PostgreSQL with instrument, owner, certificate, inspection, and audit tables
Security	Role-based auth, signed certificate payloads, encrypted storage, immutable audit events
Deployment	Managed web hosting with separate staging and production environments

7. Success measures

- Public verification completed in under 30 seconds for a known certificate.
- At least 95% of issued certificates discoverable through the public search.
- Renewal reminders sent before an instrument enters the due-soon window.
- Every status change attributable to a user, timestamp, and inspection event.
- Reduced manual reconciliation between field inspections and department registers.

8. Delivery roadmap

Phase	Deliverable
1. Foundation	Data model, authentication, roles, certificate ID scheme
2. Core portal	Registration, public lookup, certificate view, inspector queue
3. Trust layer	Digital signatures, audit log, QR-ready certificate verification
4. Pilot	One district rollout, field feedback, accessibility and security review
5. Scale	District onboarding, analytics, renewal notifications, integrations